AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271
Machine Learning and Deep Learning

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19/Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 120 Minutes**

Instructions:

- Total Marks: 30
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part – A (5 Marks)

1. Which of the following best describes the purpose of *dimensionality reduction* techniques such as PCA, Kernel PCA, and LLE?
   a) To increase the number of features in a dataset
   b) To remove irrelevant data and visualize high-dimensional data in fewer dimensions
   c) To perform clustering based on decision trees
   d) To evaluate classification performance
2. Which algorithm is limited to linear decision boundaries unless modified with kernels?
   a) Linear Regression
   b) Decision Tree
   c) Linear SVM
   d) Random Forest
3. Support Vector Machines (SVMs) can be applied to which of the following tasks?
   a) Only classification problems
   b) Only regression problems
   c) Both classification and regression problems
   d) Only clustering problems
4. Combining the outputs of several weak learners to improve predictive performance is known as:
   a) Feature selection
   b) Ensemble learning
   b) Dimensionality reduction
   d) Clustering
5. In which type of learning are all the training samples processed together rather than updated one by one?
   a. Online Learning
   b. batch Learning
   c. Reinforcement learning
   d. Incremental Learning

Part – B (15 Marks)

Provide brief answers to the following questions.

1) What is the purpose of regularization in linear models, and how do Ridge and Lasso Regression differ in their approach to reducing overfitting? (3 Marks)

2) Explain the concept of dimensionality reduction. Discuss any two techniques used for dimensionality reduction and describe how they help in improving model performance. (3 Marks)

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3) Explain how Support Vector Machines (SVMs) can perform both linear and nonlinear classification. Discuss the role of margin, support vectors, and kernel functions. (3 Marks)

4) Explain the purpose and working of a Confusion Matrix and describe how it helps in evaluating a classification model. (6 Marks)

Part – C (10 Marks)

NOTE: This question does not need any programming code.

You are working as a Data Analyst for an online movie streaming platform (similar to Netflix). The company wants to build a recommendation system that suggests movies to users based on their past ratings and viewing behavior.

Your task is to design a complete machine learning workflow to build this system.

Instructions:

- Provide a solution for building a recommendation system.
- Present your solution in a stepwise manner.
- Ensure that your response consists of a minimum of four steps.

End of questions

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